Quantify outliers

1 Identify any outliers in each of the following data sets:

a $73, 77, 81, 86, 131$

b $69, 79, 86, 72, 86, 77, 73, 82, 81, 76, 83, 47, 87, 70, 80, 85$

c $58, 63, 58, 59, 64, 68, 68, 30, 73, 25, 72, 61, 65, 69, 75, 72$

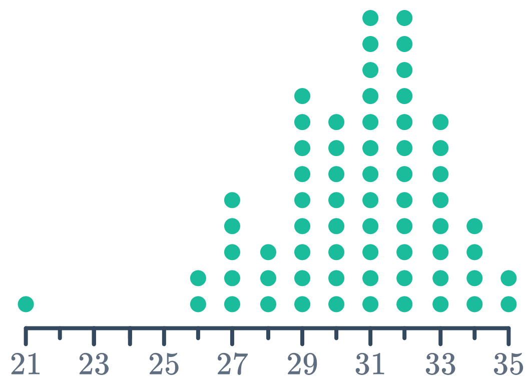
e

	Leaf
1	2
2	7 7 9 9
3	1 3 3 3 3 5 8
4	4 4 5

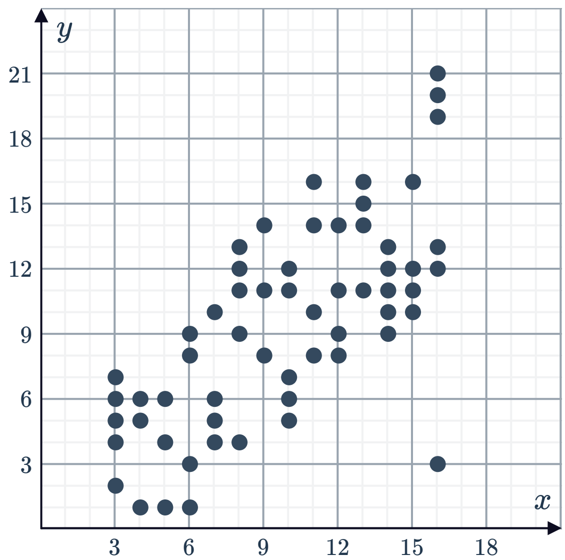
Key: $5|2 = 52$ hours

f

Temperature in a town

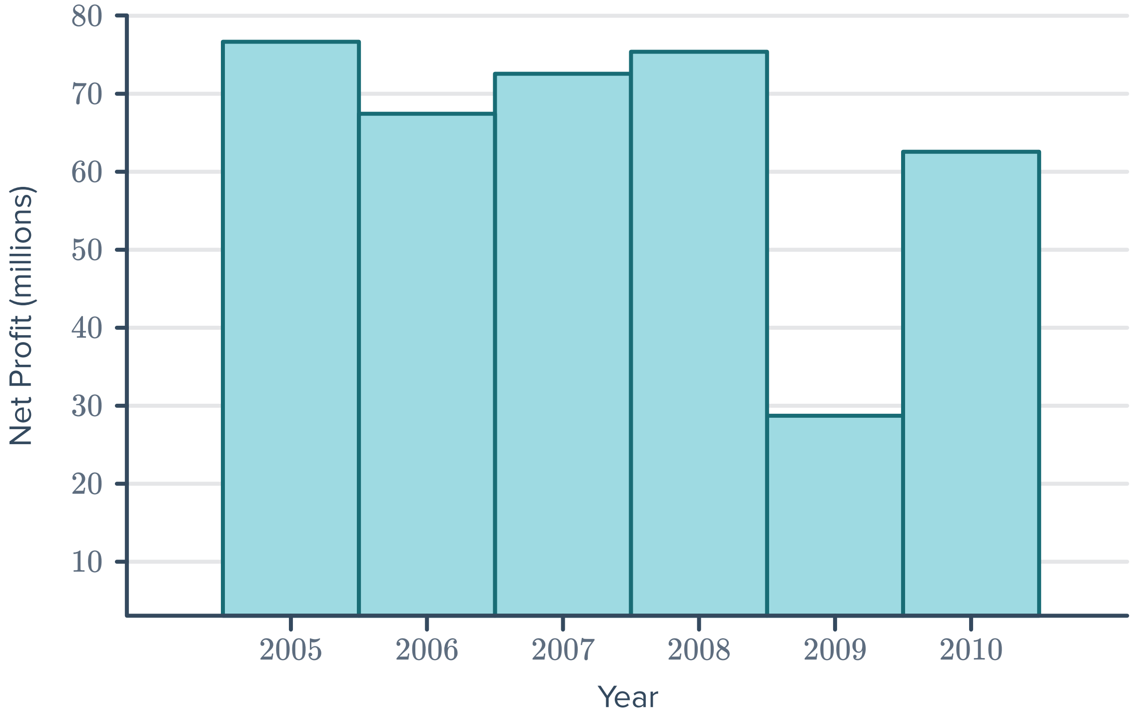


2 State the coordinates of the outlier on the following graph:

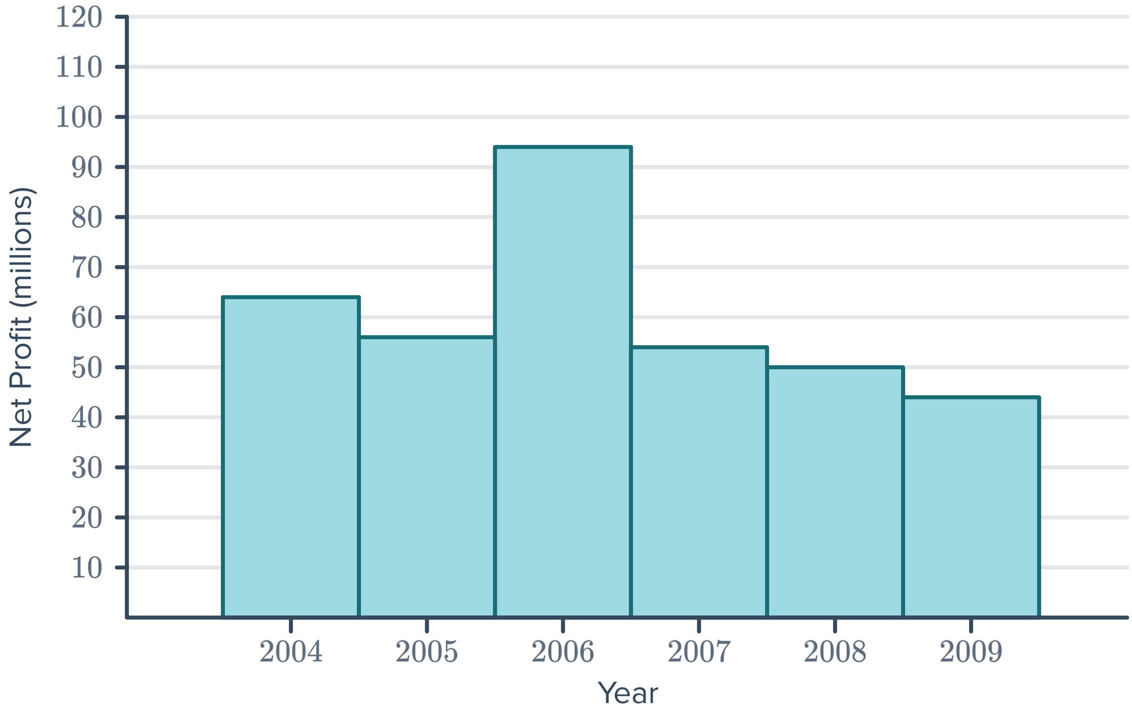


3 The following graphs show the annual net profit (in millions) of a company over a several year period. For each graph, state the year in which the annual net profit is an outlier:

a



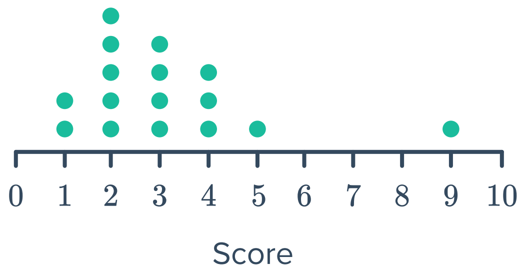
b



4 The table shows the average temperature ($^\circ\mathrm{C}$) in a particular city over several years. State the year in which the temperature is an outlier.

Year	2002	2003	2004	2005	2006	2007	2008
Temperature (°C)	31.7	26.5	22.6	22.5	24.2	23.0	24.1

5 Consider the given dot plot:



a Find the:

i Median

ii Lower quartile

iii Upper quartile

iv Interquartile range

v Value of the lower fence

vi Value of the upper fence

b Identify any outliers.

6 For each of the following data sets, calculate:

i The interquartile range

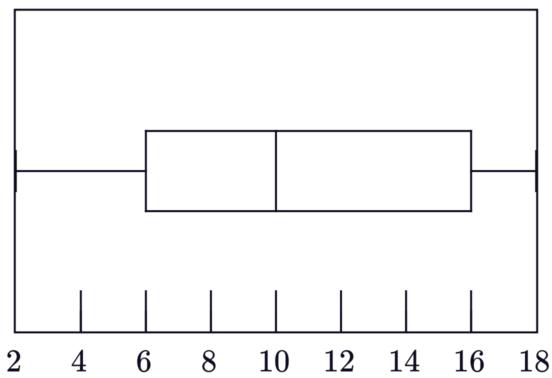
ii The value of the lower fence

iii The value of the upper fence

a

Minimum	5
Q1	6
Median	12
Q3	17
Maximum	28

b



7 For each of the following sets of data:

i Construct the five number summary.

ii Calculate the interquartile range.

iii Calculate the value of the lower fence.

iv Calculate the value of the upper fence.

v Would the value -5 be considered an outlier?

vi Would the value 16 be considered an outlier?

a $9, 5, 3, 2, 6, 1$

b $3, 10, 9, 2, 7, 5, 6$

c $12, 5, 11, 1, 9, 8, 5, 6$

8 For each of the following sets of data:

- **i** Construct the five number summary.
- **ii** Would the value -3 be considered an outlier?
- **iii** Would the value 15 be considered an outlier?

a $1, 4, 8, 10, 6, 2, 5$

b $9, 4, 6, 11, 10, 8, 10$

9 A group of Year 12 students were asked how many hours they spend on Hashtagram per day. The results are given below:

$$1.9, 1.1, 2.4, 2.3, 2.1, 1.2, 1.3, 1.6, 1.5, 1.8$$

- **a** Determine the five number summary for this data set.
- **b** Another girl, Naylaa spends 3.6 hours using Hashtagram. If her score was added to this group, would it be considered an outlier? Explain your answer.

10 The height (in metres) of certain karri trees, which grow in the south west of Australia, are shown below:

$$74, 77, 76, 81, 71, 72, 78, 75, 73, 84, 79$$

- **a** Construct the five number summary.
- **b** A tree is measured to be 66 m tall. Would this tree be considered an outlier?

11 The data point 37 is above the upper fence and is considered an outlier. The interquartile range is 10.
Find the largest integer value the upper quartile can be.

12 The data point 5 is below the lower fence and is considered an outlier. The interquartile range is 12.
Find the smallest integer value the lower quartile can be.

13 $\mathrm{VO_2}$ Max is a measure of how efficiently your body uses oxygen during exercise. The more physically fit you are, the higher your $\mathrm{VO_2}$ Max.
Here are some people's results when their $\mathrm{VO_2}$ Max was measured:

$$46,\ 27,\ 32,\ 46,\ 30,\ 25,\ 41,\ 24,\ 26,\ 29,\ 21,\ 21,\ 26,\ 47,\ 21,\ 30,\ 41,\ 26,\ 28,\ 26,\ 76$$

a Sort the values into ascending order.

b Find the median $\mathrm{VO_2}$ Max.

c Find the upper quartile value.

d Find the lower quartile value.

e Calculate $1.5 \times IQR$, where IQR is the interquartile range.

f Identify any outliers using upper and lower fences.

g Create a box plot of the data with the outlier displayed separately.

h An average untrained healthy person has a $\mathrm{VO_2}$ Max between 30 and 40.
Using the boxplot, what level of exercise is likely to describe the majority of people in this group?

Effects of outliers

14 For each of the following sets of data:

i Find the mean, median, mode, and range. Round your answers to two decimal places where necessary.

ii Identify the outlier.

iii Remove the outlier from the set and recalculate the values found in part (i).

iv Describe how each of the four statistics changed after removing the outlier.

a $53,\ 46,\ 25,\ 50,\ 30,\ 30,\ 40,\ 30,\ 47,\ 109$

b $4.7,\ 2.8,\ 1.9,\ 0.9,\ 0.9,\ 2.2,\ 2.2,\ 1.2,\ 1.5,\ 0.9$

c $4700,\ 4700,\ 4700,\ 4500,\ 5300,\ 4900,\ 5200,\ 4800,\ 1500,\ 5100$

15 For each of the following scenarios, determine whether the outlier that was removed must have had a value smaller or larger than the values that remain:

a A set of data has an outlier removed and the mean lowers.

b A set of data has an outlier removed and the mean rises.

c A set of data has an outlier removed and the median lowers.

d A set of data has an outlier removed and the median rises.

16 When an outlier is removed from a data se[redacted]cribe the effect on the following:

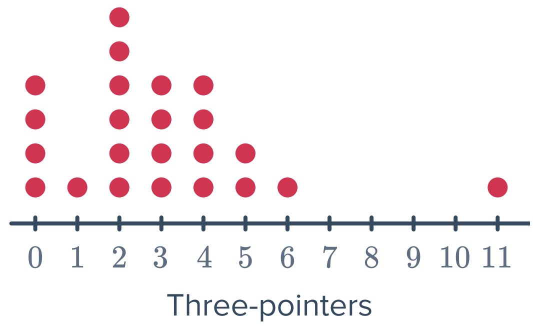
a Mode b Range c Mean d Median

17 The number of three-pointers scored in a basketball game are shown in the dot plot:
The mode is 2. If the outlier is removed, what is the new mode?

18 The number of three-pointers scored in a basketball game are shown in the dot plot:

a Find the range of the data.

b If the outlier is removed, what is the new range?



19 Consider the given stem plot:
If the outlier is removed, what is the new mean? Round your answer to two decimal places.

	Leaf
3	4 4 9
4	6 6 8 9
5	1 4
6	
7	
8	4

Key: $2|3 = 23$

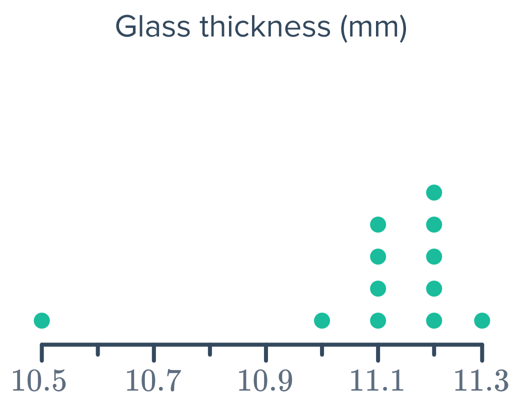
If the outlier is removed, what is the new mode?

Weight in kilograms	Frequency
14	1
15	0
16	0
17	3
18	6
19	4
20	2

21 Consider the following frequency table:
If the outlier is removed, what is the new mean? Round your answer to two decimal places if needed.

Weight in kilograms	Frequency
12	2
13	5
14	1
15	2
16	0
17	0
18	1

22 The glass windows for an airplane are rolled to a certain thickness, but machine production means there is some variation. The thickness of each pane of glass produced is measured (in millimetres), and the dot plot shows the results:



a The current median is 11.15. If the outlier is removed, what is the new median?

b The current mean is 11.1. If the outlier is removed, what is the new mean? Round your answer to two decimal places.